# CleanAlert: A Smart GPS-Enabled Real-Time Waste Collection Notification and Community Reward System for Efficient and Responsible Urban Waste Management

Devendra Singh, Sourabh Kumar, Ms. S. J. Kavitha

Department of Computer Science and Engineering, VISTAS(Vels Institute of Science And Technology), Tamilnadu, Chennai, India

**Abstract:**

In many Indian cities, household waste collection follows planned routes or fixed schedules, but residents may not know exactly when the collection vehicle will reach their street. This small timing gap can make people miss collection and leave waste outside longer than necessary. CleanAlert is proposed as an Android application that connects vehicle movement with timely information for residents. The system combines live location tracking, street-level notifications, and rewards for verified disposal. The collector location is captured using Android's FusedLocationProviderClient and displayed through Google Maps. Cloud Firestore supports real-time data synchronisation, while Firebase Cloud Messaging (FCM) delivers notifications. Residents can submit geo-tagged and timestamped photographs as disposal evidence. Verified submissions can contribute to points, daily streaks, and locality-level leaderboards. Prototype testing recorded 90% GPS street-detection accuracy (27/30 correct detections), an average notification latency of 2.82 seconds, an 88% disposal-verification rate (22/25 successful audits), and a 100% completion rate for the evaluated Login and Rewards tasks. These results provide preliminary evidence of technical feasibility under the tested conditions; broader field studies are still needed.

**Keywords:** Smart waste management; GPS tracking; geofencing; real-time notification; Firebase; gamification; mobile application; civic


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